Abishek Mosesraj

Tech Lead · Senior Python Developer · Chennai, India

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SUMMARY

Senior Python developer specialising in **backend development, RESTful APIs and GenAI**, with hands-on banking domain experience. At Thoughtworks, building a **GenAI WhatsApp banking assistant** that serves **1.2 million customers** across loans, accounts and credit cards, within the bank's data-privacy and release-governance controls. Previously Tech Lead at Roanuz, contributing to AWS infrastructure that processed over **4 billion API requests**, cutting hosting costs from **$18K to $9K per month**, and leading and mentoring a team of 5 engineers.

SKILLS

Languages	Python · SQL · JavaScript
Backend & APIs	FastAPI · Flask · Aiohttp · REST · GraphQL (Ariadne) · WebSockets · Microservices
GenAI & ML	LangGraph · CrewAI · LiteLLM · AWS Bedrock Guardrails · Langfuse · scikit-learn · PyTorch
Databases	MySQL · Redshift · Athena · MongoDB · DynamoDB · Cassandra · Redis · Elasticsearch
AWS	EC2 · S3 · Lambda · ECS · Fargate · API Gateway · SQS · SNS · EventBridge · CDK · CloudWatch
Data & Messaging	Apache Kafka · RabbitMQ · Celery · Event-Driven Architecture · Pandas · PyArrow
DevOps & CI/CD	Docker · Kubernetes (basic) · GitHub Actions · AWS CodePipeline · Git
Testing	Pytest
AI Tools	Claude Code

EXPERIENCE

Thoughtworks - Client: Enterprise Client

Senior Consultant

Jun 2025 - Present · Chennai

- Core team member on a **bilingual GenAI banking assistant** on WhatsApp and web, serving **1.2M customers**.
- Feature owner for **shared platform features** in a **29-member Agile/Scrum team**, contributing to the in-house LangGraph workflow engine.
- Own **intent classification** for the assistant, along with its monitoring, evaluation and release process.

Roanuz

Tech Lead

Dec 2021 - May 2025 · Chennai

- Designed AWS infrastructure on EC2, S3, Lambda and ECS, contributing to platforms that served **4B+ API requests**.
- Led hosting cost optimisation, reducing infrastructure spend from **$18K to $9K/month (50% reduction)**.
- Led a team of **5 engineers** building scalable Python-based sports data management and analytics platforms.
- Built a real-time collaborative sports editing platform using **WebSockets**, improving team productivity and data accuracy.
- Implemented **Docker and AWS ECS** containerisation, speeding up deployments.
- Established CI/CD pipelines using **AWS CodePipeline**; served as Developer Advocate for internal framework.

Software Engineer

May 2019 - Dec 2021 · Chennai

- Developed and optimised **REST APIs** for sports analytics platforms handling **50,000 requests/hour** at average load.
- Set up AWS cloud infrastructure for cricket and football data, designed to scale from day one.
- Improved IT Operations Support with proper workload tracking, leading to the establishment of a dedicated IT Ops team.
- Recognised as **Standout Performer 2021** for technical excellence and process improvements.

Amazon

ML Data Associate

Nov 2016 - May 2019 · Chennai

- Built **automation tools** for data workflow bottlenecks, speeding up delivery.
- Drove QA quality improvements from **88% to 92%** through systematic error pattern analysis.
- Received the **Extra Mile Award** for continuous contributions to process improvement.

KEY PROJECTS

GenAI Banking Assistant

Senior Consultant · Thoughtworks

24/7 bilingual (English and Hindi) AI customer-service assistant on WhatsApp and web for a retail bank, serving **1.2M customers**.

- Self-service for **~45 journeys** across loans, cards and accounts, such as card block, loan foreclosure and disputes.
- Understands everyday language (Hinglish, typos, short replies), offering "Did you mean...?" options when unsure.
- Handles multiple requests per chat, follow-ups, corrections and **human handoff**.
- Secure by design: verification, **MFA**, PII protection and AI safety checks; new journeys are added through configuration.

Tech: Python · FastAPI · LangGraph · CrewAI · LiteLLM · AWS Bedrock Guardrails · scikit-learn · PyTorch · MongoDB · Redis · Kafka · Langfuse

API Billing & Monetisation System ↗

Software Engineer · Roanuz

Usage-based billing system for CricketAPI's sports data APIs, with plan selection, usage tracking and payment gateway integration.

- Integrated **Stripe** and **Razorpay** payment gateways using AWS API Gateway, EventBridge and Lambda.
- Built usage-tracking APIs, a billing transparency dashboard and email alerts for plan limits.
- Set up **Envoy Proxy** to route requests to different servers by path prefix.

Tech: Python · Flask · Nginx · Envoy · DocumentDB · Memcached · Stripe · Razorpay

Real-time Collaborative Sports Editor (ANTS) ↗

Tech Lead (Team of 5) · Roanuz

Led 5 developers to deliver a real-time cricket editing platform that processes live match data with sub-second latency.

- Optimised scorecard building with **Apache Kafka**, making data processing more efficient.
- Set code quality standards through PR reviews; coordinated deployment across 5 EC2 servers and 4 AWS managed services.
- Managed the move from the legacy editor with **minimal downtime**.

Tech: Python · Kafka · RabbitMQ · DynamoDB · Cassandra · Meilisearch · Aiohttp

Internal Framework Enhancements

Developer Advocate · Roanuz

Extended Roanuz's internal Python application framework with built-in search, message queue processing and package management.

- Integrated **Elasticsearch** and built serverless **Fargate** task orchestration for RabbitMQ, Kafka and SQS messages.
- Set up an **AWS CodeArtifact** publishing pipeline to streamline dependency management.
- Onboarded and mentored **2 teams (6 developers)** onto the framework and drove adoption across apps.

Tech: Python · AWS CDK · Elasticsearch · Fargate · CodeArtifact

EDUCATION

B.E. Mechanical Engineering

May 2016

Alpha College of Engineering · Chennai